



**TK2023**

## **Object-Oriented Software Engineering**

**UKMFolio:**

### **A University Learning Management System**

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# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 INTRODUCTION**

The UKMFOLIO project was initiated to address the growing need for an integrated, efficient platform that supports the academic activities of UKM (Universiti Kebangsaan Malaysia). With the shift towards digital learning, both lecturers and students have been challenged by the difficulties of managing and accessing course content, submitting assignments, tracking progress, and staying up-to-date with course announcements. Existing systems often fall short, requiring the use of multiple tools that can lead to inefficiencies and a disjointed experience. The goal of UKMFOLIO is to bridge this gap by providing a centralized, user-friendly platform that brings together all essential academic functions into one cohesive system. This project aims to simplify administrative tasks, enhance communication between lecturers and students, and create an engaging learning environment that fosters student success and supports lecturers in their teaching.

### **1.2 PROBLEM STATEMENT**

Managing online learning resources and student progress can be a complex and time-consuming task for both lecturers and students. For lecturers, keeping up with the administration of multiple courses means juggling various tools for uploading materials, creating assignments, and grading submissions. This fragmented process often leads to inefficiencies, lost time, and increased stress. The challenge of manually tracking students' progress and maintaining clear communication adds to the workload, detracting from the time that could be spent on meaningful teaching and engagement. From the students' perspective, the current systems can be equally difficult to navigate. They often need to access different platforms to view course materials, submit assignments, and check their grades, which can lead to confusion and missed deadlines. The lack of an integrated system makes it harder for students to stay organized, track their progress, and communicate effectively with their lecturers and peers. These challenges underscore the need for a more streamlined and user-friendly solution that simplifies academic processes for everyone involved.

### **1.3 PROPOSED SOLUTION**

UKMFOLIO is a system that provides a centralized platform for management of courses, quizzes, and assignments, communication between lecturers and students, and tracking student performance. The system allows lecturers to grade assignments, create quizzes, upload course materials and send announcements to students, all in one place. Meanwhile, students can hand in their assignments, view and download course materials, answer quizzes and check their progress on each course on the very same platform. This solves the problem of having to grade papers and keep track of assignments manually, both for lecturers and students.

The intended users of this software are lecturers and students of a university. The features of UKMFolio include Course Material Management, Forum Discussions, Assignments, Quizzes and Announcements. Course Material Management refers to the function for lecturers to upload course materials such as lecture notes and references, and for students to view said materials and download them. Meanwhile, Forum Discussions are a feature where lecturers are able to start or post a forum discussion about a certain topic and students are able to view the forum and comment under the post. Lecturers can also create Assignments, which is a page for students to submit their work by uploading file(s) to be reviewed and graded by the lecturer. Lecturers can also provide feedback in a comment section on the Assignment page. For the Quiz feature, lecturers can set up the questions and answers of a quiz in UKMFOLIO and post it in their course menu, and when students answer the quiz, their answers will be graded automatically based on the answers set by the lecturer. Finally, the Announcement feature is for lecturers to notify students about certain updates or information. When a lecturer posts an announcement, students receive a notification in their notification menu, and can be redirected to the announcement page by clicking on it. The announcements are also visible on the course menu itself, within a submenu named “Announcements”.

Within our team, the people responsible for the development of each functionality are as follows:

| <b>Name</b>                   | <b>Functionality</b>           |
|-------------------------------|--------------------------------|
| Eva Ray Siew                  | Course Material Management     |
| Evelyn Foo Yifei              | Forum Discussions, Assignments |
| Evelyn Ng Zi Wei              | Assignments                    |
| Ng Zhen Xuan                  | Log in, Log out, Quizzes       |
| Syamil Syah Danial bin Fadzil | Announcements                  |

## **CHAPTER 2**

### **SOFTWARE REQUIREMENTS**

#### **2.1 INTRODUCTION**

In order to build a successful system, it is essential to clearly define both the functional and non-functional software requirements. These requirements are essential for the development process, as they outline what the platform needs to do and how it needs to perform in real-world conditions. The goal is to ensure that UKMFOLIO meets the needs of its primary users such as lecturers and students while also being efficient, secure, and adaptable enough to evolve as the university's needs change over time.

To achieve this, the UKMFOLIO platform must provide a robust set of features, including course material management, assignment submission and grading, quizzes, and announcements. In addition to these core functions, the system needs to be scalable, accessible, and reliable, offering users a consistent and high-quality experience. Furthermore, the platform must be secure, protecting sensitive academic data and user privacy.

As we move through the development process, this chapter will outline the key software requirements that will guide the development of the UKMFOLIO system. These requirements are divided into functional and system requirements, ensuring that both the user-facing features and the technical aspects of the system are addressed. In addition, we will also discuss the quality standards the platform must meet to ensure a positive user experience and the constraints that may impact the development process, such as time, budget, and technical limitations.

By identifying and addressing these key requirements, the UKMFOLIO platform will be positioned to enhance the academic experience at UKM, streamlining administrative tasks and improving communication between lecturers and students. This comprehensive approach will not only simplify the management of academic content but also foster a more engaging and efficient learning environment.

## **2.2 FUNCTIONAL REQUIREMENT**

### 2.2.1 USER REQUIREMENTS

#### A. Lecturers' Requirements:

1. **Course Management:** Lecturers shall be able to create and manage multiple courses from a single dashboard. This includes uploading lecture notes, assignment, guidelines, and other course materials.
2. **Assignment Management:** Lecturers shall be able to create assignments, set deadlines, and grade student submissions. The system shall allow them to provide personalized feedback on assignments.
3. **Quiz Management:** The system shall enable lecturers to design quizzes with multiple options, automatically grade responses, and offer instant feedback to students.
4. **Communication Tools:** Lecturers shall be able to post announcements, initiate forum discussions, and interact with students through messages or discussions.
5. **Progress Tracking:** Lecturers need a reliable way to track student progress across assignments, quizzes, and overall performance, with detailed reporting options for individual students.

#### B. Students' Requirements:

1. **Course Access:** Students shall be able to easily access all their enrolled courses, including course materials, assignments, quizzes, and announcements from a centralized location.
2. **Assignment Submission:** Students need a straightforward method to upload their assignments. The platform shall automatically confirm submissions and display deadlines.
3. **Quiz Participation:** Students shall be able to take quizzes, receive immediate feedback, and view their scores after submission.
4. **Forum Interaction:** Students shall be able to read, comment on, and participate in forum discussions, creating an interactive environment with their peers and lecturers.
5. **Progress Dashboard:** Students shall have access to a personal dashboard that provides a quick overview of their academic progress, including assignment scores, quiz results, and upcoming deadlines.

### 2.2.2 SYSTEM REQUIREMENTS

## 2.3 QUALITY REQUIREMENTS

1. Usability:
  - a. The system shall be easy for everyone to use regardless of their skills.
  - b. Both lecturers and students shall be able to navigate the platform without needing a lot of training or support.
  - c. The interface shall be intuitive and clear, so users can focus on their academic tasks, not on figuring out how the system works.
2. Performance:
  - a. The platform shall be fast and responsive.
  - b. Pages shall load quickly, and students and lecturers shall be able to access course materials, submit assignments, and take quizzes without delay.
  - c. During busy times (like right before assignments are due), the system needs to handle many users without slowing down.
3. Reliability:
  - a. The system shall work consistently, with minimal downtime.
  - b. Students and lecturers shall be able to rely on the platform to be available when they need it.
  - c. Any outages shall be rare and, if they happen, resolved quickly. The platform shall have backup systems in place so that if something goes wrong, there's no loss of important data.
4. Security:
  - a. Given the sensitive nature of the data involved (like grades, assignments, and personal information), the system shall be secure. This includes using strong passwords, encrypting data, and ensuring that only authorized users can access their accounts and materials.
  - b. The platform shall be regularly checked for vulnerabilities to keep it safe from potential threats.
5. Maintainability:
  - a. The system shall be easy to maintain and update. As new features or improvements are needed, the platform shall be flexible enough to accommodate them without causing major disruptions.
  - b. The development team shall be able to troubleshoot issues quickly, update software, and keep everything running smoothly without too much effort.

## 2.4 CONSTRAINTS

While developing UKMFOLIO, there are certain constraints or limitations that we need to keep in mind. These could come from various areas like time, budget, technology, or other external factors. Recognizing and understanding these constraints helps set realistic expectations and ensures we can deliver a system that meets everyone's needs, even if there are some challenges along the way.

1. Time Constraints:
  - a. The project has a set timeline that we need to stick to. This means certain features might need to be prioritized over others, and we may need to delay some advanced functionalities for future updates. The platform shall be ready by the start of the academic year, so development, testing, and user training all need to happen on a tight schedule.
2. Budget Constraints:
  - a. There is a budget allocated for the development of the system, and this could limit how much we can do initially. For example, the choice of technology or the size of the development team might be affected by the budget. We also need to be mindful of costs for infrastructure, cloud hosting, or additional tools that might be needed.
3. Technical Constraints:
  - a. The platform needs to be compatible with the university's existing systems, such as the student database and email systems. This could limit some of the design decisions or require additional integration work. Moreover, we will also be working with the technology stack that the team is already familiar with, which might influence the tools and programming languages used.
4. User Adoption:
  - a. While the platform is designed to make academic processes easier, there may be some resistance to change, especially from users who are used to the current systems. Lecturers and students might need time to adjust, and additional training and support will be necessary to help them get the most out of UKMFOLIO.

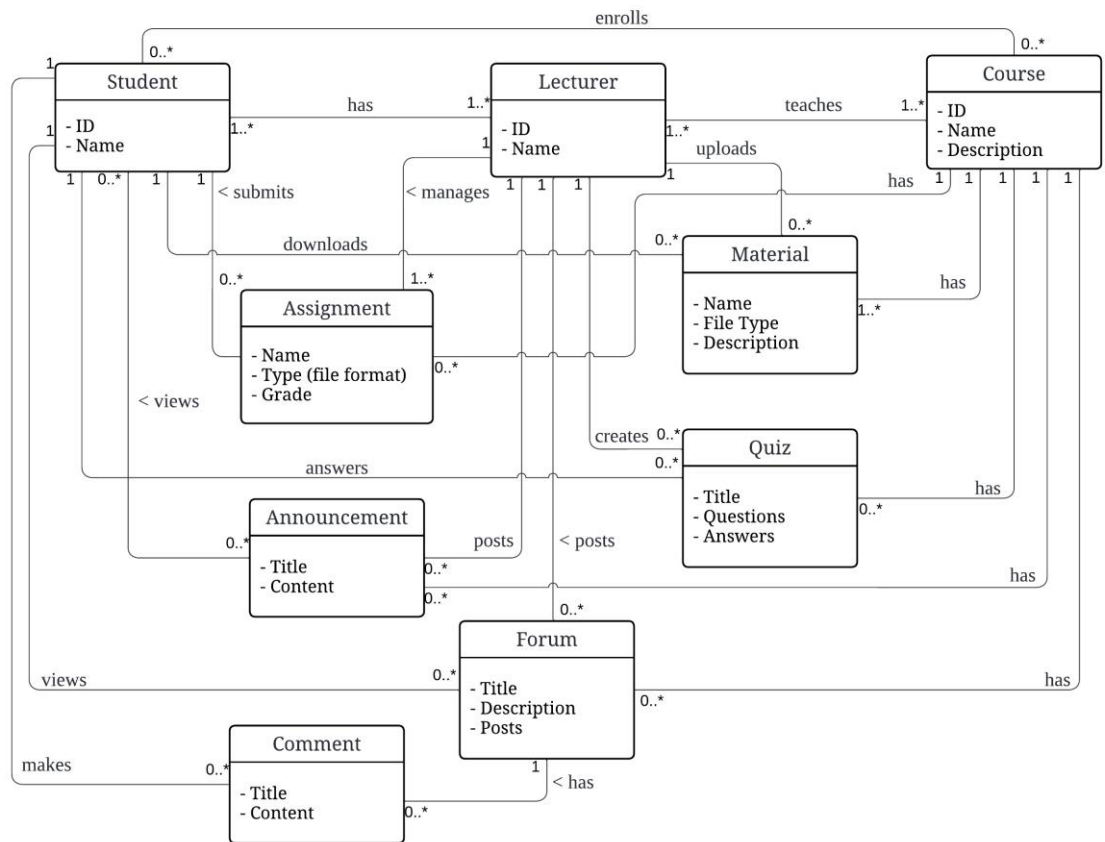
5. Legal and Compliance Constraints:

- a. The system shall comply with relevant laws and regulations around data protection and privacy, especially as it deals with student records and personal information. This means following best practices for data security and adhering to local laws, which may impact how we store, access, and share user data.

6. Resource Availability:

- a. The availability of resources, like skilled developers and technical support staff, could also be a limiting factor. If key team members are unavailable or if there are other competing priorities, it could impact the timeline or quality of the final product. Additionally, the university's IT infrastructure needs to be up to the task of hosting and maintaining the platform.

## 2.5 DOMAIN MODEL



**Figure 1** Domain Model of UKMFolio

The domain model diagram for UKMFolio defines the key entities within the Learning Management System (LMS) and their relationships. In the Learning management system, there is a User entity that generalizes two main actors: Lecturer and Student, and this is the main component of the system. Referred to as the user, whose attributes include user ID, name, email, and password. And, a lecturer manages courses and is tasked with uploading materials, creating assignments and quizzes, grading submissions, and posting announcements. The system's other part, the Student, engages with the system by enrolling into courses, viewing instructional materials, submitting work, taking quizzes, and reading announcements.

A lecturer creates and manages each course, which is the central focal entity of all academic peripherals. In a course, lecturers are also enabled to upload Course Materials which consists of files such as lecture notes, slides, and reference documents to be provided to the students for their learning. These materials can also be seen and downloaded by the learners. Furthermore, teachers are able to set Assignments where

learners are required to turn in their work before the cut off date. Each assignment is linked to one or more Submissions, where students upload their responses. The submissions will be reviewed and graded by the lecturers, then feedback will be provided.

The system features a Quiz module which signifies assessments created by the lecturers. These assessments comprise multiple choice questions with set answers and may also have a timer. Students are allowed to attempt these assessments which are automatically scored by the system. After marking, the grades are stored and made available for students and lecturers to monitor.

In order to communicate, the Announcement entity enables lecturers to upload critical information, updates, or anything relevant to the course. After uploading an announcement, students that registered for the course will be able to see it on their dashboard, allowing them to be aware of any changes or forthcoming deadlines.

These relations will create a flow in the system within the LMS. Each lecturer holds complete power in managing courses while the students interact with the course by using the materials, doing exercises, and taking the exams. This domain model explains how UKMFolio functions by providing and maintaining clear representation in regard to students and lecturers for their interaction with every component of the system to make their learning experience simple and straightforward.

## 2.6 USE CASE

### 2.6.1 USE CASE DIAGRAM



The use case diagram in the document illustrates the interactions between the system and its primary users, which include lecturers and students. It provides a high-level overview of the functionalities available within UKMFolio and how each user engages with the system. The diagram consists of various use cases, representing distinct actions that users can perform, along with the relationships between these actions and the actors involved.

Lecturers have multiple responsibilities within the system. They can create and manage courses, allowing them to upload course materials, post announcements, and manage assignments and quizzes. The ability to upload course materials enables lecturers to

share lecture notes, references, and other learning resources with students. They can also create and manage assignments by setting deadlines, reviewing submissions, and providing grades and feedback. Additionally, lecturers can create quizzes, which students can attempt within a specified time frame. Once a quiz is completed, the system automatically grades it and provides results. Another essential function lecturers perform is posting announcements, which serve as a means of communication to inform students about important updates related to their courses.

Students interact with the system in various ways as well. They can access their enrolled courses, view and download course materials, and participate in discussions through forum posts. Students are also responsible for submitting assignments before the deadline, and after submission, they can view their grades and lecturer feedback. Additionally, students can take quizzes to assess their knowledge and receive immediate feedback on their performance. They also have access to announcements posted by lecturers, ensuring they remain informed about any course-related updates.

The use case diagram effectively captures the interactions between users and the system, emphasizing how lecturers and students engage with UKMFolio to enhance learning and course management. By visualizing these interactions, the diagram provides a clear understanding of the system's capabilities and ensures that all essential academic activities are streamlined within a single platform.

## 2.6.2 USE CASE SPECIFICATIONS

|                        |                                                                                               |                                                           |
|------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ID:                    | UC-01                                                                                         |                                                           |
| Title:                 | Create and Manage Quizzes                                                                     |                                                           |
| Description:           | Lecturer creates, edits, and deletes quizzes for student to attempt, and view student's marks |                                                           |
| Primary Actor:         | Lecturer                                                                                      |                                                           |
| Precondition:          | Lecturer must be logged into the system                                                       |                                                           |
| Postcondition:         | Quiz is successfully created, edited or deleted.                                              |                                                           |
| Main Success Scenario: | Actor                                                                                         | System                                                    |
|                        | 1. Lecturer clicks add quiz button.                                                           |                                                           |
|                        |                                                                                               | 2. System prompts for quiz details.                       |
|                        | 3. Lecturer enters quiz title and questions.                                                  |                                                           |
|                        |                                                                                               | 4. System saves quiz and confirms creation.               |
|                        |                                                                                               | 5. System displays created quiz for selection.            |
|                        | 6. Lecturer selects a quiz and click "Edit Quiz" button                                       |                                                           |
|                        |                                                                                               | 7. System preload selected quiz details                   |
|                        | 8. Lecturer modifies details                                                                  |                                                           |
|                        |                                                                                               | 9. System updates quiz and saves changes                  |
|                        | 10. Lecturer selects a quiz and click "Delete Quiz" button                                    |                                                           |
|                        |                                                                                               | 11. System removes selected quiz after confirmation.      |
|                        | 12. Lecturer selects a quiz and click "View Student Marks" button                             |                                                           |
|                        |                                                                                               | 13. System display marks for all registered student       |
| Alternative Scenarios: | Actor                                                                                         | System                                                    |
|                        | 3a. Lecturer enters incomplete quiz details                                                   |                                                           |
|                        |                                                                                               | 4a. System prompts for missing details before proceeding. |

|  |                                          |                                          |
|--|------------------------------------------|------------------------------------------|
|  | 6a., 12a. Lecturer did not select a quiz |                                          |
|  |                                          | 7a., 13a. System displays error message. |
|  | 10a. Lecturer cancels deletion.          |                                          |
|  |                                          | 11a. System retains the quiz.            |

|                        |                                                            |                                                                                                                |
|------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| ID:                    | UC-02                                                      |                                                                                                                |
| Title:                 | Attempt Quiz and View Scores                               |                                                                                                                |
| Description:           | Student selects a quiz, answers questions, and view scores |                                                                                                                |
| Primary Actor:         | Student                                                    |                                                                                                                |
| Precondition:          | Student must be logged into the system                     |                                                                                                                |
| Postcondition:         | Quiz answers are submitted, and scores are stored          |                                                                                                                |
| Main Success Scenario: | Actor                                                      | System                                                                                                         |
|                        |                                                            | 1. System displays available quizzes.                                                                          |
|                        | 2. Student selects a quiz and click “Start Quiz” button    |                                                                                                                |
|                        |                                                            | 3. System loads quiz questions.                                                                                |
|                        | 4. Student answers questions                               |                                                                                                                |
|                        |                                                            | 5. System records answers                                                                                      |
|                        | 6. Student submits quiz                                    |                                                                                                                |
|                        |                                                            | 7. System calculates score and store student answer and score.                                                 |
|                        | 8. Student selects a quiz and click “View Quiz” button     |                                                                                                                |
|                        |                                                            | 9. System loads quiz with submitted answers. If an answer is incorrect the system displays the correct answer. |
|                        |                                                            | 10. System displays the total marks obtained.                                                                  |
| Alternative Scenarios: | Actor                                                      | System                                                                                                         |
|                        | 2a1. Student did not select a quiz                         |                                                                                                                |
|                        | 2a2. Student submitted the selected quiz before.           |                                                                                                                |
|                        |                                                            | 3a1. System displays an error message.                                                                         |
|                        |                                                            | 3a2. System displays an error message                                                                          |

|  |                                          |                                             |
|--|------------------------------------------|---------------------------------------------|
|  | 6a. Students submits and incomplete quiz |                                             |
|  |                                          | 7a. System wants about unanswered question. |
|  | 8a. Student selects a not submitted quiz |                                             |
|  |                                          | 9a. System displays an error message.       |

|                        |                                                                                              |                                                                                                 |
|------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| ID:                    | UC-03                                                                                        |                                                                                                 |
| Title:                 | Create and Manage Forums                                                                     |                                                                                                 |
| Description:           | Lecturer creates Forum Discussions in which students can view its content and post comments. |                                                                                                 |
| Primary Actor:         | Lecturer                                                                                     |                                                                                                 |
| Precondition:          | Lecturer must be logged into the system                                                      |                                                                                                 |
| Postcondition:         | Forum is successfully created and displayed to the user.                                     |                                                                                                 |
| Main Success Scenario: | Actor                                                                                        | System                                                                                          |
|                        | 1. Lecturer enters forum topic and content and presses “Create Discussion” button            |                                                                                                 |
|                        |                                                                                              | 2. System displays a success message and displays the newly added forum topic in the menu below |
|                        | 3. Lecturer selects a forum discussion from the topic menu                                   |                                                                                                 |
|                        |                                                                                              | 4. System displays the content of the selected forum and comments (if exists)                   |

|                        |                                                 |                                      |
|------------------------|-------------------------------------------------|--------------------------------------|
| ID:                    | UC-04                                           |                                      |
| Title:                 | View Forum Discussions and Leave Comments       |                                      |
| Description:           | Students view and comment on forum discussions. |                                      |
| Primary Actor:         | Student                                         |                                      |
| Precondition:          | Student must be logged into the system          |                                      |
| Postcondition:         | Comment(s) is/are posted                        |                                      |
| Main Success Scenario: | Actor                                           | System                               |
|                        | 1. Student selects a forum topic from the list  |                                      |
|                        |                                                 | 2. System displays the forum content |

|  |                                                           |                                                    |
|--|-----------------------------------------------------------|----------------------------------------------------|
|  | 3. Student enters comment and clicks “Add Comment” button |                                                    |
|  |                                                           | 4. System displays new comment under forum content |

|                        |                                                                              |                                                                 |
|------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------|
| ID:                    | UC-05                                                                        |                                                                 |
| Title:                 | View Announcements                                                           |                                                                 |
| Description:           | Student selects an announcement created by a lecturer, and views the details |                                                                 |
| Primary Actor:         | Student                                                                      |                                                                 |
| Precondition:          | Student must be logged into the system                                       |                                                                 |
| Postcondition:         | Announcement must be created by lecturer                                     |                                                                 |
| Main Success Scenario: | Actor                                                                        | System                                                          |
|                        |                                                                              | 1. System displays available announcements created by lecturer. |
|                        | 2. Student selects an announcement.                                          |                                                                 |
|                        |                                                                              | 3. System loads and displays the announcement chosen.           |
|                        | 4. Student views the announcement                                            |                                                                 |

|                        |                                                                                                           |                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| ID:                    | UC-06                                                                                                     |                                               |
| Title:                 | Create and Edit Announcements                                                                             |                                               |
| Description:           | Lecturer creates and edit Announcements in which students can view its content.                           |                                               |
| Primary Actor:         | Lecturer                                                                                                  |                                               |
| Precondition:          | Lecturer must be logged into the system                                                                   |                                               |
| Postcondition:         | Announcement is successfully created or updated by lecturer.                                              |                                               |
| Main Success Scenario: | Actor                                                                                                     | System                                        |
|                        | 1. Lecturer navigates to the "Announcements" section                                                      |                                               |
|                        |                                                                                                           | 2. System load and displays announcement tab. |
|                        | 3. Lecturer clicks on "Add Announcement" to create a new announcement or selects an existing one to edit. |                                               |

|                        |                                                                                           |                                                                       |
|------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|                        |                                                                                           | 4. System creates a new blank announcement for lecturer to input.     |
|                        |                                                                                           | 5. System loads and displays the selected announcement.               |
|                        | 6. Lecturer enters or modifies the announcement details, including the title and content. |                                                                       |
|                        | 7. Lecturer confirms and saves the announcement.                                          |                                                                       |
|                        |                                                                                           | 9. System updates and saves the list of announcements.                |
|                        |                                                                                           | 10. The updated announcement is displayed for students.               |
| Alternative Scenarios: | Actor                                                                                     | System                                                                |
|                        | 3a. Lecturer leave blank fields when submitting.                                          |                                                                       |
|                        |                                                                                           | 4a. System displays error message regarding blank field.              |
|                        |                                                                                           | 9a. System fails to save announcement                                 |
|                        |                                                                                           | 10a. System displays error message and prompts lecturer to try again. |

|                        |                                                                                                                                                         |        |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| ID:                    | UC-07                                                                                                                                                   |        |
| Title:                 | Upload Course Materials                                                                                                                                 |        |
| Description:           | Lecturer uploads course materials to be accessed by students.                                                                                           |        |
| Primary Actor:         | Lecturer                                                                                                                                                |        |
| Precondition:          | <ul style="list-style-type: none"> <li>Lecturer must be logged into the system.</li> <li>Lecturer has the course materials ready for upload.</li> </ul> |        |
| Postcondition:         | The course materials are uploaded and available for students to view and download.                                                                      |        |
| Main Success Scenario: | Actor                                                                                                                                                   | System |
|                        | 1. Lecturer selects the course and topic.                                                                                                               |        |
|                        | 2. Lecturer adds materials using the “Add Material” button.                                                                                             |        |
|                        | 3. Lecturer uploads the materials using the “Upload Materials” button.                                                                                  |        |

|                        |                                                                                                               |                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                               | 4. System saves the materials and updates the student view.                                                           |
| Alternative Scenarios: | Actor                                                                                                         | System                                                                                                                |
|                        | 3a1. Lecturer cancels the file selection dialog.                                                              |                                                                                                                       |
|                        |                                                                                                               | 3a2. System does not add any material to the list.                                                                    |
|                        | 3a3. Lecturer can attempt to add materials again.                                                             |                                                                                                                       |
|                        | 4a1. Lecturer attempts to upload materials without filling in all the fields or without adding any materials. |                                                                                                                       |
|                        |                                                                                                               | 4a2. System shows an error message indicating that all fields must be filled and at least one material must be added. |
|                        | 4a3. Lecturer corrects the input and attempts to upload again.                                                |                                                                                                                       |
|                        |                                                                                                               | 4b1. System encounters an error while saving the materials.                                                           |
|                        |                                                                                                               | 4b2. System shows an error message indicating the failure.                                                            |
|                        | 4b3. Lecturer can attempt to upload again.                                                                    |                                                                                                                       |

|                        |                                                                                                                                                      |        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| ID:                    | UC-08                                                                                                                                                |        |
| Title:                 | View Course Materials                                                                                                                                |        |
| Description:           | Student views and downloads course materials uploaded by the lecturer.                                                                               |        |
| Primary Actor:         | Student                                                                                                                                              |        |
| Precondition:          | <ul style="list-style-type: none"> <li>Student must be logged into the system.</li> <li>Lecturer must have uploaded the course materials.</li> </ul> |        |
| Postcondition:         | The student views or downloads the selected course materials.                                                                                        |        |
| Main Success Scenario: | Actor                                                                                                                                                | System |
|                        | 1. Student selects the course and topic.                                                                                                             |        |
|                        | 2. Student views the list of available materials.                                                                                                    |        |
|                        | 3. Student selects a material and clicks the “View Material” button.                                                                                 |        |

|                        |                                                                              |                                                                                                           |
|------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|                        |                                                                              | 4. System displays the material content.                                                                  |
|                        | 5. Student has the option to download the material.                          |                                                                                                           |
| Alternative Scenarios: | Actor                                                                        | System                                                                                                    |
|                        | 2a1. Student selects a course and topic with no available materials.         |                                                                                                           |
|                        |                                                                              | 2a2. System shows a message indicating that no materials are available for the selected course and topic. |
|                        | 3a1. Student clicks the “View Material” button without selecting a material. |                                                                                                           |
|                        |                                                                              | 3a2. System shows an error message indicating that a material must be selected to view its content.       |
|                        | 4a1. Student clicks the “Download” button.                                   |                                                                                                           |
|                        |                                                                              | 4a2. System encounters an error while attempting to download the material.                                |
|                        |                                                                              | 4a3. System shows an error message indicating the failure.                                                |
|                        | 4a4. Student can attempt to download again.                                  |                                                                                                           |

|                        |                                                                             |        |
|------------------------|-----------------------------------------------------------------------------|--------|
| ID:                    | UC-09                                                                       |        |
| Title:                 | Create Assignment                                                           |        |
| Description:           | Lecturer creates a new assignment for the students.                         |        |
| Primary Actor:         | Lecturer                                                                    |        |
| Precondition:          | Lecturer must be logged into the system.                                    |        |
| Postcondition:         | The new assignment is created and visible to students.                      |        |
| Main Success Scenario: | Actor                                                                       | System |
|                        | 1. Lecturer navigates to the assignment creation form.                      |        |
|                        | 2. Lecturer fills in the assignment details (title, description, due date). |        |
|                        | 3. Lecturer submits the form.                                               |        |

|                        |                                                              |                                                                     |
|------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|
|                        |                                                              | 4. System saves the assignment and updates the list of assignments. |
| Alternative Scenarios: | Actor                                                        | System                                                              |
|                        | 2a1. Lecturer submits the form with missing or invalid data. |                                                                     |
|                        |                                                              | 2a2. System shows an error message indicating the required fields.  |
|                        | 2a3. Lecturer corrects the input and submits the form again. |                                                                     |

|                        |                                                                                                                                                               |                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| ID:                    | UC-10                                                                                                                                                         |                                         |
| Title:                 | Grade Submission                                                                                                                                              |                                         |
| Description:           | Lecturer grades a student's submission.                                                                                                                       |                                         |
| Primary Actor:         | Lecturer                                                                                                                                                      |                                         |
| Precondition:          | <ul style="list-style-type: none"> <li>Lecturer must be logged into the system.</li> <li>There must be at least one submission for the assignment.</li> </ul> |                                         |
| Postcondition:         | The student's submission is graded and the grade is saved.                                                                                                    |                                         |
| Main Success Scenario: | Actor                                                                                                                                                         | System                                  |
|                        | 1. Lecturer selects an assignment.                                                                                                                            |                                         |
|                        | 2. Lecturer views the list of submissions.                                                                                                                    |                                         |
|                        | 3. Lecturer selects a submission to grade.                                                                                                                    |                                         |
|                        | 4. Lecturer enters the grade and feedback.                                                                                                                    |                                         |
|                        | 5. Lecturer submits the grade.                                                                                                                                |                                         |
|                        |                                                                                                                                                               | 6. System saves the grade and feedback. |
| Alternative Scenarios: | Actor                                                                                                                                                         | System                                  |
|                        | 4a1. Lecturer attempts to grade a submission that has already been graded.                                                                                    |                                         |
|                        |                                                                                                                                                               | 4a2. System shows a warning message.    |
|                        | 4a3. Lecturer can choose to update the grade or leave it unchanged.                                                                                           |                                         |

|        |                   |  |
|--------|-------------------|--|
| ID:    | UC-11             |  |
| Title: | Upload Submission |  |

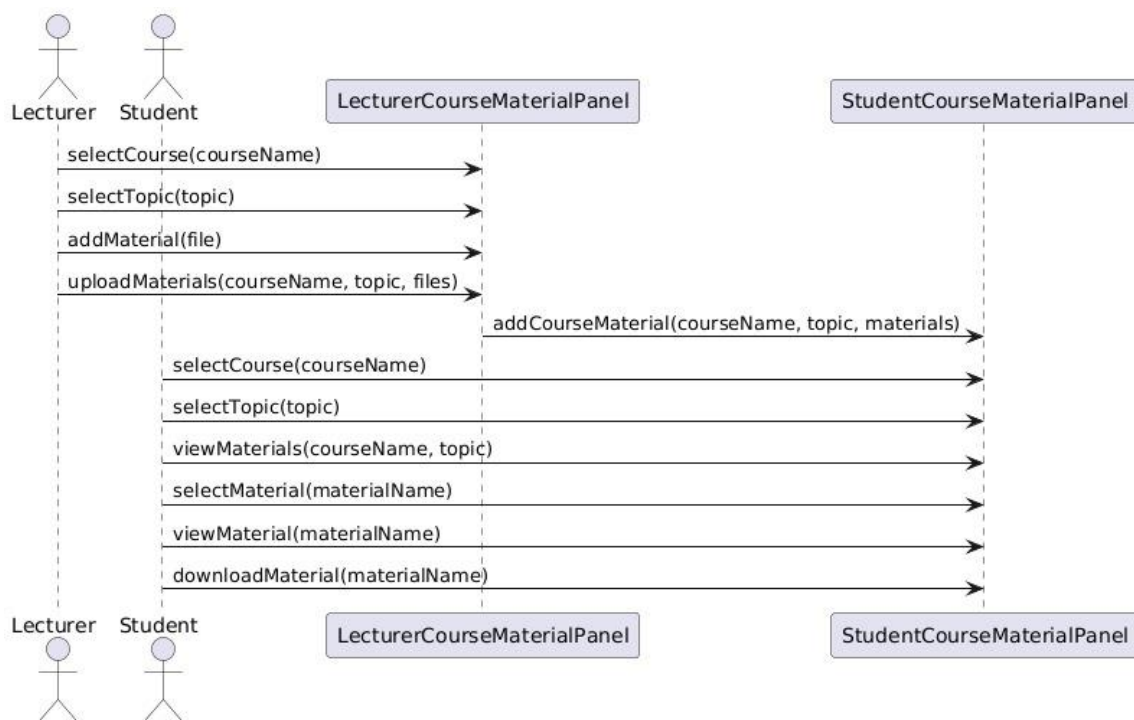
|                        |                                                                                                                                        |                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Description:           | Student uploads a submission for an assignment.                                                                                        |                                                                   |
| Primary Actor:         | Student                                                                                                                                |                                                                   |
| Precondition:          | <ul style="list-style-type: none"> <li>• Student must be logged in.</li> <li>• The assignment must be open for submissions.</li> </ul> |                                                                   |
| Postcondition:         | The submission is uploaded and saved.                                                                                                  |                                                                   |
| Main Success Scenario: | Actor                                                                                                                                  | System                                                            |
|                        | 1. Student selects an assignment.                                                                                                      |                                                                   |
|                        | 2. Student clicks the “Upload Submission” button.                                                                                      |                                                                   |
|                        | 3. Student selects a file to upload.                                                                                                   |                                                                   |
|                        | 4. Student confirms the upload.                                                                                                        |                                                                   |
|                        |                                                                                                                                        | 5. System saves the submission and updates the assignment status. |
| Alternative Scenarios: | Actor                                                                                                                                  | System                                                            |
|                        | 3a1. Student cancels the file selection dialog.                                                                                        |                                                                   |
|                        |                                                                                                                                        | 3a2. System does not upload any file.                             |
|                        | 4a1. Student has already submitted for the assignment.                                                                                 |                                                                   |
|                        |                                                                                                                                        | 4a2. System shows a warning message.                              |
|                        | 4a3. Student can choose to replace the existing submission or cancel the upload.                                                       |                                                                   |

|                        |                                                                                                                      |        |
|------------------------|----------------------------------------------------------------------------------------------------------------------|--------|
| ID:                    | UC-12                                                                                                                |        |
| Title:                 | View Assignment Details                                                                                              |        |
| Description:           | Student views the details of an assignment, including any submission made.                                           |        |
| Primary Actor:         | Student                                                                                                              |        |
| Precondition:          | <ul style="list-style-type: none"> <li>• Student must be logged in.</li> <li>• The assignment must exist.</li> </ul> |        |
| Postcondition:         | The student views the assignment details.                                                                            |        |
| Main Success Scenario: | Actor                                                                                                                | System |
|                        | 1. Student selects an assignment.                                                                                    |        |
|                        | 2. Student clicks the “View Details” button.                                                                         |        |

|                        |                                                             |                                                                                              |
|------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                        |                                                             | 5. System displays the assignment details, including any submitted file, grade and feedback. |
| Alternative Scenarios: | Actor                                                       | System                                                                                       |
|                        | 2a1. Student has not made any submission for the assignment |                                                                                              |
|                        |                                                             | 2a2. System shows the assignment details without submission-specific information.            |

### 2.6.3 SYSTEM SEQUENCE DIAGRAMS

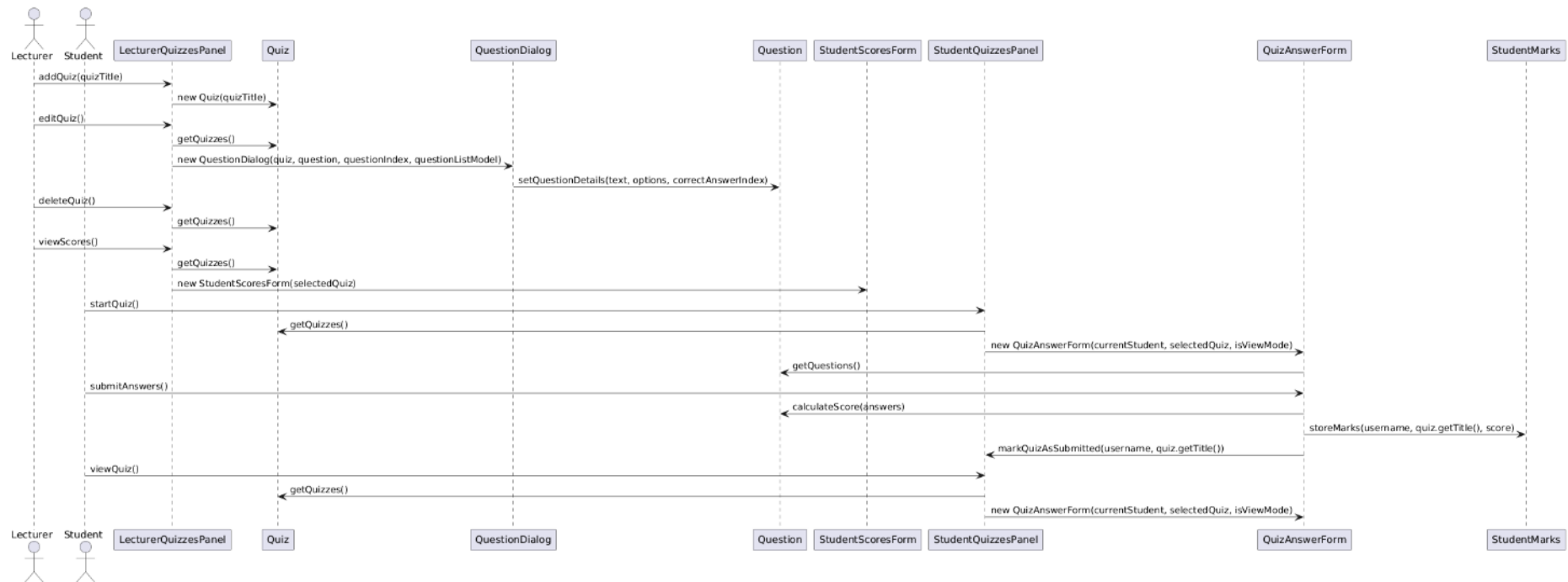
#### 1. Course Material



The SSD diagram illustrates the interaction between lecturers and students with the course material management system. The process begins with a lecturer selecting a course and a specific topic within that course using the **LecturerCourseMaterialPanel**. The lecturer then adds new material by uploading files related to the chosen course and topic. Once the materials are uploaded, the **LecturerCourseMaterialPanel** communicates with the **StudentCourseMaterialPanel** to update the course materials available for students. On the student side, the interaction starts with selecting a course and then a topic within that course using the **StudentCourseMaterialPanel**. The student can view the list of available materials for the selected course and topic. When a student selects a specific material, they can either view

the content directly or download it for offline use. The system ensures that both lecturers and students have an efficient and organized way to manage and access course materials.

## 2. Quiz

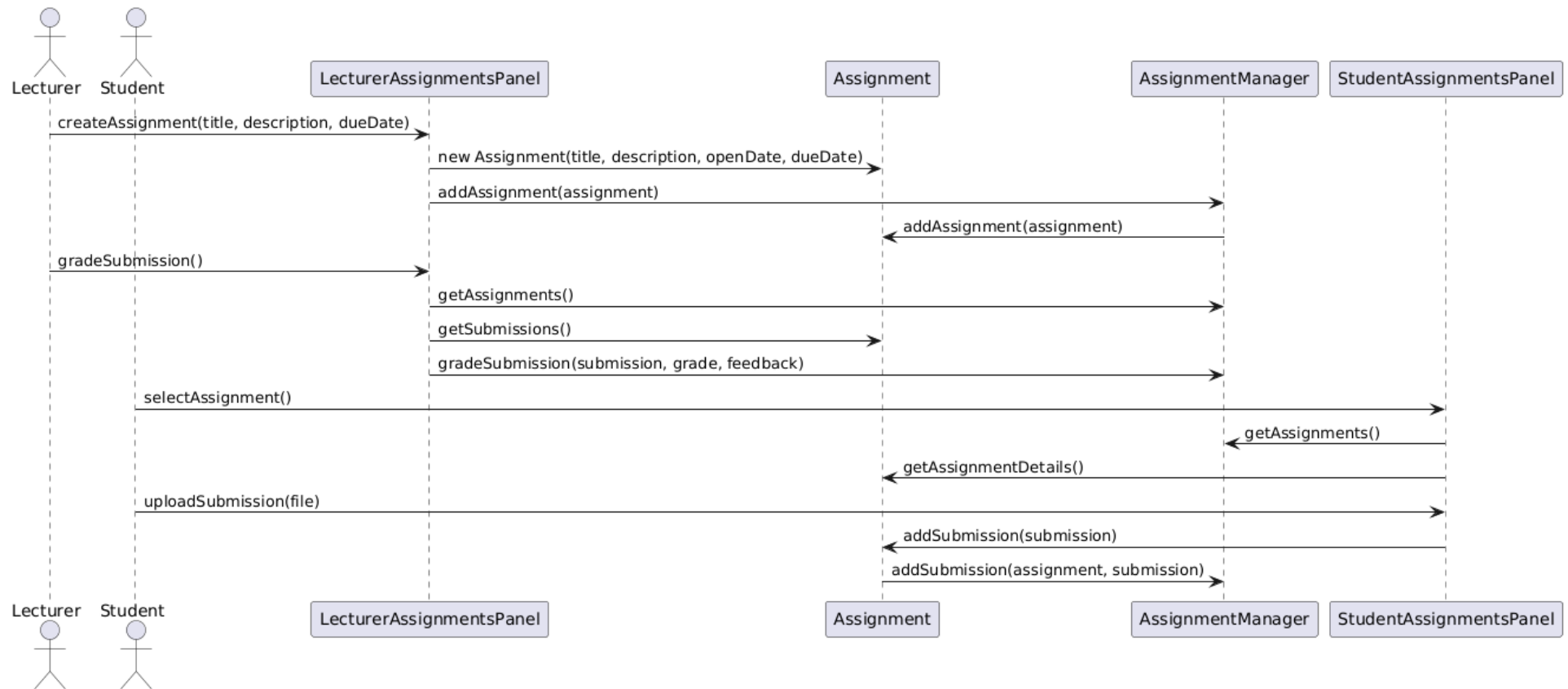


The SSD diagram illustrates the interaction between lecturers and students with the quiz management system. The process begins with a lecturer adding a new quiz by using the LecturerQuizzesPanel to create a new Quiz object with a specified title. The lecturer can then edit the quiz by retrieving the list of quizzes and using the QuestionDialog to set the details of individual questions, including the text, options, and correct answer. If necessary, the lecturer can delete a quiz by retrieving the list of quizzes and selecting the one to remove. Additionally, the lecturer can view student scores for a specific quiz by retrieving the list of quizzes and generating a new StudentScoresForm for the selected quiz.

On the student side, the interaction starts with the student using the StudentQuizzesPanel to start a quiz. The panel retrieves the list of available quizzes and initializes a QuizAnswerForm for the selected quiz in either quiz-taking mode or view mode. The QuizAnswerForm fetches the relevant questions for the quiz. When the student submits their answers, the form calculates the score based on the provided answers, stores the

marks associated with the student's username, and updates the StudentQuizzesPanel to mark the quiz as submitted. Students can also view a completed quiz by selecting it from the list of quizzes and initializing the QuizAnswerForm in view mode. The system ensures an organized and efficient way for both lecturers and students to manage and participate in quizzes.

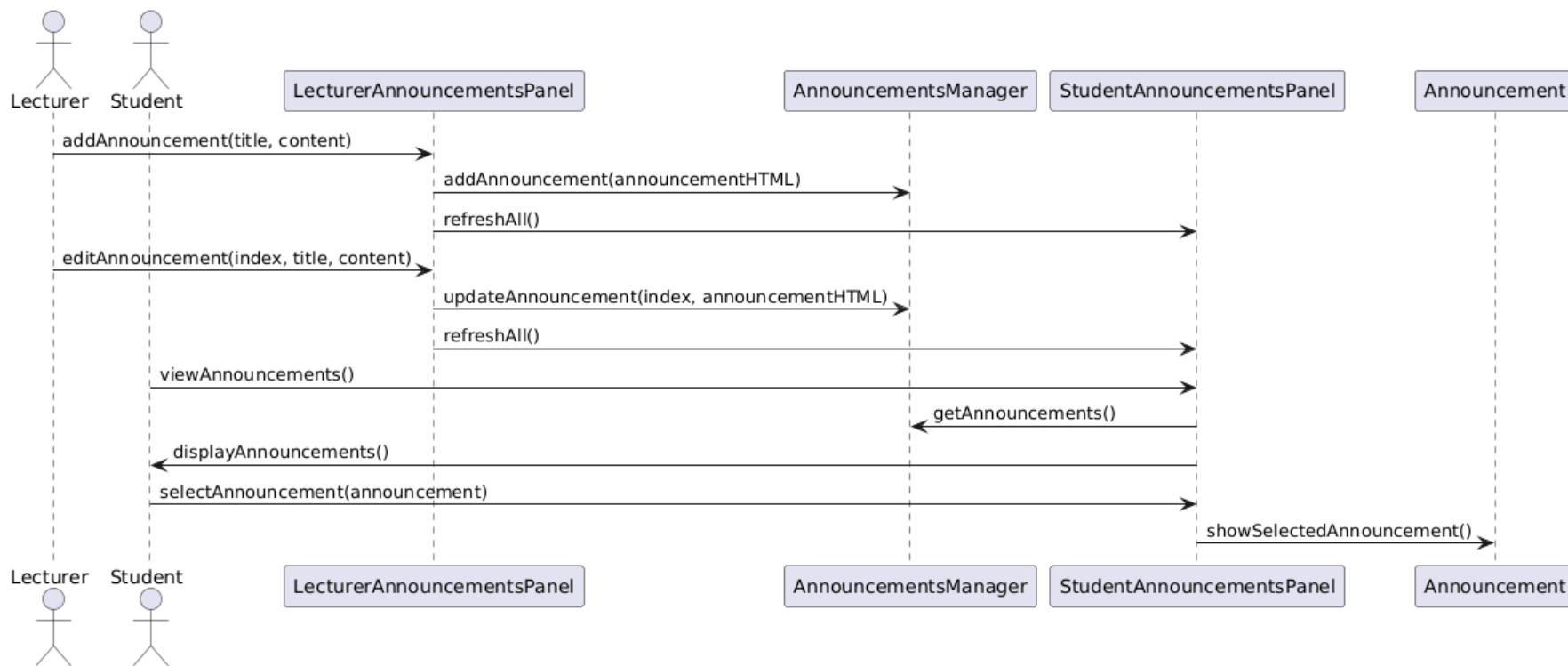
### 3. Assignments



The SSD diagram illustrates the interaction between lecturers and students with the assignment management system. The process begins with a lecturer creating a new assignment by using the LecturerAssignmentsPanel to provide the assignment title, description, and due date. This action creates a new Assignment object, which is then added to the AssignmentManager and subsequently stored. When a lecturer needs to grade a submission, they use the LecturerAssignmentsPanel to retrieve the list of assignments from the AssignmentManager, obtain the submissions for a specific assignment, and then grade a particular submission by providing the grade and feedback to the AssignmentManager.

On the student side, the interaction starts with the student using the StudentAssignmentsPanel to select an assignment. The panel retrieves the list of available assignments from the AssignmentManager and displays the details of the selected assignment. When a student uploads a submission, the StudentAssignmentsPanel sends the submission to the corresponding Assignment object, which then forwards it to the AssignmentManager for storage. The system ensures a structured and efficient way for both lecturers and students to manage and participate in assignment-related activities.

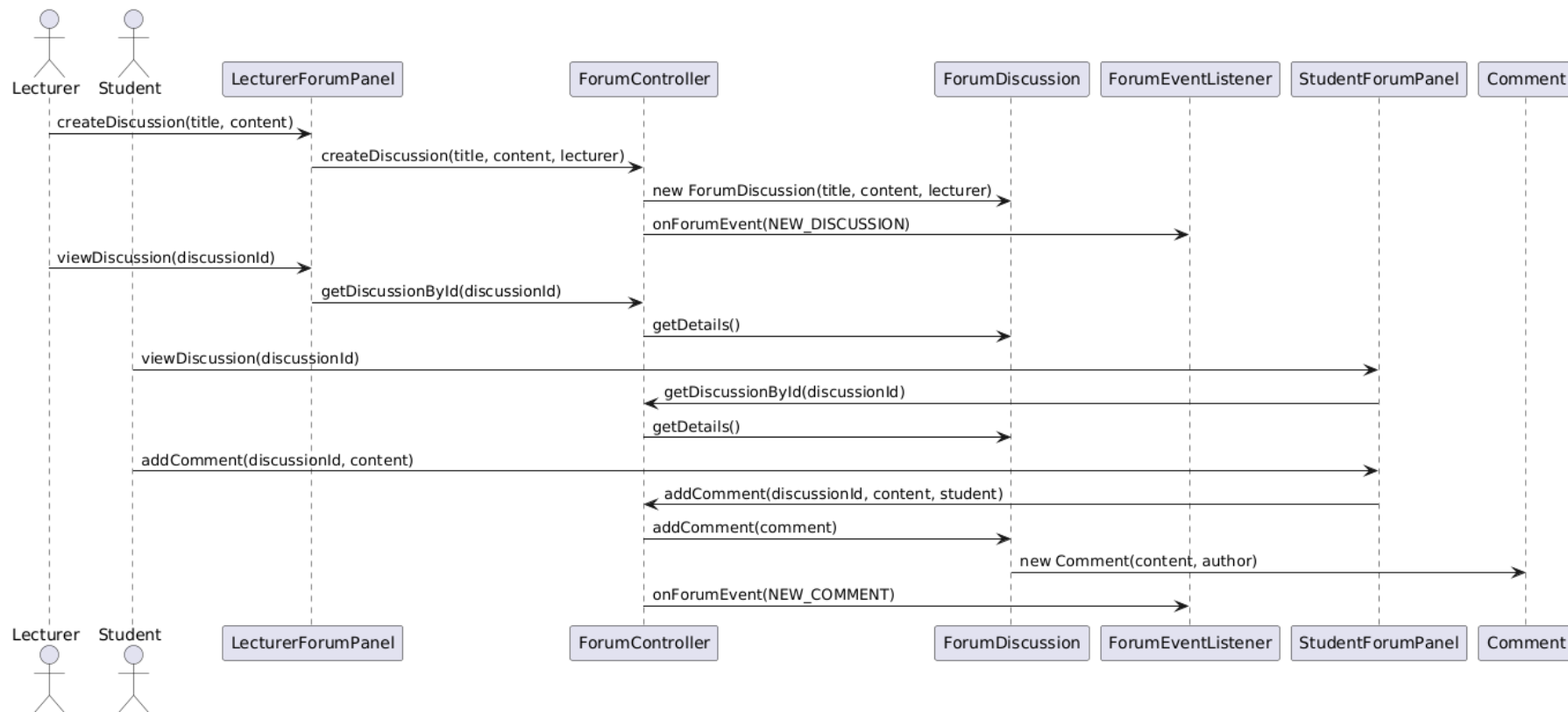
#### 4. Announcements



The SSD diagram illustrates the interaction between lecturers, students, and the announcement management system. The process begins with a lecturer adding a new announcement through the LecturerAnnouncementsPanel, which sends the announcement's title and content to the AnnouncementsManager in HTML format. After the announcement is added, the LecturerAnnouncementsPanel triggers the StudentAnnouncementsPanel to refresh and display the updated list of announcements. If the lecturer needs to edit an existing announcement, they use the LecturerAnnouncementsPanel to provide the updated title and content, specifying the index of the announcement to be edited. The AnnouncementsManager updates the announcement, and the LecturerAnnouncementsPanel once again prompts the StudentAnnouncementsPanel to refresh the list.

On the student side, the interaction starts with the student viewing announcements using the `StudentAnnouncementsPanel`, which retrieves the list of announcements from the `AnnouncementsManager` and displays them to the student. When a student selects a specific announcement, the `StudentAnnouncementsPanel` shows the detailed content of that announcement through the `Announcement` component. The system ensures a structured and efficient way for both lecturers and students to manage and interact with announcements.

## 5. Forum



The SSD diagram illustrates the interaction between lecturers, students, and the forum management system. The process begins with a lecturer creating a new forum discussion using the `LecturerForumPanel`, which sends the discussion title and content to the `ForumController`. The `ForumController` then creates a new `ForumDiscussion` object with the provided details and notifies the `ForumEventListener` of the new discussion event. Lecturers can also view a specific discussion by selecting it from the `LecturerForumPanel`, which retrieves the discussion details from the `ForumController` and the associated `ForumDiscussion` object.

On the student side, the interaction starts with students viewing discussions using the `StudentForumPanel`, which functions similarly by retrieving discussion details from the `ForumController` and the `ForumDiscussion` object. Students can also add comments to a discussion by providing the

discussion ID and comment content to the StudentForumPanel, which forwards the information to the ForumController. The ForumController then adds the comment to the relevant ForumDiscussion object, creates a new Comment object with the provided content and author details, and notifies the ForumEventListener of the new comment event. The system ensures a seamless and organized way for both lecturers and students to manage and participate in forum discussions.

## **CHAPTER 3**

### **DESIGN AND IMPLEMENTATION**

#### **3.1 INTRODUCTION**

Course management systems such as UKMFolio need to be properly designed and implemented in order to guarantee that the system efficiently meets the user's expectations. UKMFolio is a system that aims to provide a centralized platform for management of courses, quizzes, and assignments, communication between lecturers and students, and tracking student performance. The system allows lecturers to grade assignments, create quizzes, upload course materials and send announcements to students, all in one place. Meanwhile, students can hand in their assignments, view and download course materials, answer quizzes and check their progress on each course on the very same platform. This chapter mainly focuses on the overall design of the software and implementing it to the system.

### 3.2 CLASS DIAGRAM

| Class Diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="208 328 577 853"> <div>MainFrame</div> <div> -JFrame frame<br/> -JPanel cardPanel<br/> -CardLayout cardLayout<br/> -RolePanel rolePanel<br/> -LoginPage loginPage<br/> -UserRegistration userRegistration<br/> -String loggedInUsername </div> <div> +MainFrame()<br/> +showRegistration()<br/> +showLogin()<br/> +showDashboard(String role)<br/> +setLoggedInUsername(String)<br/> +getLoggedInUsername()<br/> +main(String[] args) </div> </div> | <p><b>Attributes</b></p> <ul style="list-style-type: none"> <li>- frame: JFrame</li> <li>- cardPanel: JPanel</li> <li>- cardLayout: CardLayout</li> <li>- rolePanel: RolePanel</li> <li>- loginPanel: LoginPage</li> <li>- userRegistration: UserRegistration</li> <li>- loggedInUsername: String</li> </ul> <p><b>Methods</b></p> <ul style="list-style-type: none"> <li>- MainFrame(): Constructor</li> <li>- showRegistration(): void</li> <li>- showLogin(): void</li> <li>- showDashboard(String role): void</li> <li>- setLoggedInUsername(String username): void</li> <li>- getLoggedInUsername(): String</li> </ul> |
| <div data-bbox="208 906 577 1220"> <div>LoginPage</div> <div> -JTextField usernameField<br/> -JPasswordField passwordField<br/> -MainFrame mainFrame </div> <div> +LoginPage(MainFrame)<br/> -loginUser()<br/> +clearFields() </div> </div>                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="206 226 577 547"> <div>UserRegistration</div> <div> -JTextField usernameField<br/> -JPasswordField passwordField<br/> -JComboBox roleBox<br/> -MainFrame mainFrame </div> <div> +UserRegistration(MainFrame)<br/> +registerUser()<br/> +clearFields() </div> </div>                                                                                                                                                      |  |
| <div data-bbox="206 603 577 1129"> <div>User</div> <div> -String username<br/> -String password<br/> -String role<br/> -static ArrayList&lt;User&gt; users<br/> -static User currentUser </div> <div> +User(String, String, String)<br/> +getUsername()<br/> +getPassword()<br/> +getRole()<br/> +static getUsers()<br/> +static getUserByUsername(String)<br/> +static getCurrentUser()<br/> +static setCurrentUser(User) </div> </div> |  |

|                                                                                                                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div> <div>RolePanel</div> <div> #JPanel menuPanel<br/> #JPanel contentPanel<br/> -String userRole<br/> -CardLayout cardLayout<br/> -MainFrame mainFrame </div> <div>+RolePanel(MainFrame, String)</div> </div> |  |
| <div> <div>StudentPanel</div> <div> -String username<br/> -List&lt;Quiz&gt; quizzes </div> <div>+StudentPanel(MainFrame, String)</div> </div>                                                                   |  |
| <div> <div>LecturerPanel</div> <div></div> <div>+LecturerPanel(MainFrame, String)</div> </div>                                                                                                                  |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div>LecturerForumPanel</div> <div><div><div>-ForumController controller</div><div>-JList&lt;ForumDiscussion&gt; discussionList</div><div>-DefaultListModel&lt;ForumDiscussion&gt; discussionListModel</div><div>-JPanel discussionPanel</div><div>-JTextArea contentArea</div><div>-ForumDiscussion currentDiscussion</div><div>-SimpleDateFormat dateFormat</div></div><div><div>+LecturerForumPanel()</div><div>+onForumEvent(ForumEvent)</div><div>-initializeUI()</div><div>-updateDiscussionList()</div><div>-updateDiscussionDisplay()</div><div>-createCommentBox(Comment)</div></div></div> |  |
| <div>StudentForumPanel</div> <div><div><div>-ForumController controller</div><div>-JList&lt;ForumDiscussion&gt; discussionList</div><div>-DefaultListModel&lt;ForumDiscussion&gt; discussionListModel</div><div>-JPanel discussionPanel</div><div>-ForumDiscussion currentDiscussion</div><div>-SimpleDateFormat dateFormat</div></div><div><div>+StudentForumPanel()</div><div>-initializeUI()</div><div>-updateDiscussionList()</div><div>-updateDiscussionDisplay()</div><div>-createCommentBox(Comment)</div><div>+onForumEvent(ForumEvent)</div></div></div>                                    |  |

| <table><tr><th>ForumController</th></tr><tr><td>-static ForumController instance<br/>-ArrayList&lt;ForumDiscussion&gt; discussions<br/>-List&lt;ForumEventListener&gt; listeners</td></tr><tr><td>-ForumController()<br/>+static getInstance()<br/>+addEventListener(ForumEventListener)<br/>+removeEventListener(ForumEventListener)<br/>-notifyListeners(String, Object)<br/>+createDiscussion(String, String, String)<br/>+addComment(int, String, String)<br/>+getDiscussions()<br/>+getDiscussionById(int)</td></tr></table> | ForumController | -static ForumController instance<br>-ArrayList<ForumDiscussion> discussions<br>-List<ForumEventListener> listeners                       | -ForumController()<br>+static getInstance()<br>+addEventListener(ForumEventListener)<br>+removeEventListener(ForumEventListener)<br>-notifyListeners(String, Object)<br>+createDiscussion(String, String, String)<br>+addComment(int, String, String)<br>+getDiscussions()<br>+getDiscussionById(int) |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| ForumController                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |
| -static ForumController instance<br>-ArrayList<ForumDiscussion> discussions<br>-List<ForumEventListener> listeners                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |
| -ForumController()<br>+static getInstance()<br>+addEventListener(ForumEventListener)<br>+removeEventListener(ForumEventListener)<br>-notifyListeners(String, Object)<br>+createDiscussion(String, String, String)<br>+addComment(int, String, String)<br>+getDiscussions()<br>+getDiscussionById(int)                                                                                                                                                                                                                             |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |
| <table><tr><th>ForumDiscussion</th></tr><tr><td>-int id<br/>-String title<br/>-String content<br/>-String lecturer<br/>-Date timestamp<br/>-ArrayList&lt;Comment&gt; comments<br/>-static int nextId</td></tr><tr><td>+ForumDiscussion(String, String, String)<br/>+getId()<br/>+getTitle()<br/>+getContent()<br/>+getLecturer()<br/>+getTimestamp()<br/>+getComments()<br/>+addComment(Comment)</td></tr></table>                                                                                                                | ForumDiscussion | -int id<br>-String title<br>-String content<br>-String lecturer<br>-Date timestamp<br>-ArrayList<Comment> comments<br>-static int nextId | +ForumDiscussion(String, String, String)<br>+getId()<br>+getTitle()<br>+getContent()<br>+getLecturer()<br>+getTimestamp()<br>+getComments()<br>+addComment(Comment)                                                                                                                                   |  |
| ForumDiscussion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |
| -int id<br>-String title<br>-String content<br>-String lecturer<br>-Date timestamp<br>-ArrayList<Comment> comments<br>-static int nextId                                                                                                                                                                                                                                                                                                                                                                                          |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |
| +ForumDiscussion(String, String, String)<br>+getId()<br>+getTitle()<br>+getContent()<br>+getLecturer()<br>+getTimestamp()<br>+getComments()<br>+addComment(Comment)                                                                                                                                                                                                                                                                                                                                                               |                 |                                                                                                                                          |                                                                                                                                                                                                                                                                                                       |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 236 564 520"> <div>ForumEvent</div> <div>           -String type<br/>           -Object data         </div> <div>           +ForumEvent(String, Object)<br/>           +getType()<br/>           +getData()         </div> </div>                                                                                                                                                                                               |  |
| <div data-bbox="208 579 622 753"> <div>ForumEventListener</div> <div></div> <div>           +onForumEvent(ForumEvent)         </div> </div>                                                                                                                                                                                                                                                                                                         |  |
| <div data-bbox="208 813 530 1268"> <div>Comment</div> <div>           -Integer id<br/>           -String content<br/>           -String author<br/>           -Date timestamp<br/>           -static Integer nextid         </div> <div>           +Comment(String,String)<br/>           +getId() : Integer<br/>           +getContent() : String<br/>           +getAuthor() : String<br/>           +getTimeStamp() : Date         </div> </div> |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 225 714 625"> <div>StudentCourseMaterialPanel</div> <div> -JComboBox courseComboBox<br/> -JComboBox topicComboBox<br/> -JList materialList<br/> -DefaultListModel materialListModel<br/> -static Map&lt;&gt; courseMaterials </div> <div> +StudentCourseMaterialPanel()<br/> +static addCourseMaterial(String, String, List)<br/> -updateCourses()<br/> -updateTopics(String)<br/> -updateMaterials(String, String)<br/> -showMaterialPopup(String) </div> </div> |  |
| <div data-bbox="208 679 696 1037"> <div>LecturerCourseMaterialPanel</div> <div> -JTextField courseNameField<br/> -JTextField topicField<br/> -JButton addMaterialButton<br/> -JButton uploadButton<br/> -JList materialList<br/> -DefaultListModel materialListModel<br/> -List materials </div> <div>+LecturerCourseMaterialPanel()</div> </div>                                                                                                                                     |  |

| LecturerAssignmentsPanel                                                                                                                                                                                                                                                                                                                                                                      |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| -AssignmentManager manager<br>-JTable assignmentsTable<br>-DefaultTableModel tableModel<br>-JButton createButton<br>-JButton gradeButton<br>-JButton viewSubmissionsButton                                                                                                                                                                                                                    |  |
| +LecturerAssignmentsPanel()<br>-initializeUI()<br>-viewSubmissions()<br>-showSubmissionsDialog(Assignment assignment, List<Submission> submissions)<br>-showFileContentDialog(Submission submission)<br>-setupListeners()<br>-updateButtonStates()<br>-refreshTable()<br>-showCreateDialog()<br>-gradeSubmission()<br>-showGradingDialog(Assignment assignment, List<Submission> submissions) |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 225 689 782"> <div>StudentAssignmentsPanel</div> <div> -AssignmentManager manager<br/> -JTable assignmentsTable<br/> -JButton uploadButton<br/> -JButton viewDetailsButton<br/> -JLabel statusLabel<br/> -DefaultTableModel tableModel<br/> -Assignment selectedAssignment </div> <div> +StudentAssignmentsPanel()<br/> -initializeUI()<br/> -setupListeners()<br/> -updateButtonStates(boolean)<br/> -updateStatusLabel(Assignment)<br/> -uploadSubmission()<br/> -viewAssignmentDetails()<br/> -addDetailRow(JPanel, GridBagConstraints, String, String)<br/> -refreshTable()<br/> -handleError(String, Exception)<br/> -refreshPanel() </div> </div> |  |
| <div data-bbox="208 831 808 1201"> <div>AssignmentManager</div> <div> -static AssignmentManager instance<br/> -List&lt;Assignment&gt; assignments<br/> -MapUnsupported markdown: del&gt; submissionsMap </div> <div> -AssignmentManager()<br/> +static getInstance() : AssignmentManager<br/> +addAssignment(Assignment assignment)<br/> +addSubmission(Assignment assignment, Submission submission)<br/> +getSubmissions(Assignment assignment) : List&lt;Submission&gt;<br/> +gradeSubmission(Submission submission, int grade, String feedback)<br/> +getAssignments() : List&lt;Assignment&gt; </div> </div>                                                           |  |

| Assignment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div><div>-String title</div><div>-String description</div><div>-LocalDateTime submissionOpenDate</div><div>-LocalDateTime submissionDeadline</div><div>-File assignmentFile</div><div>-List&lt;Submission&gt; submissions</div><div>-boolean isOpen</div><div>-String status</div></div>                                                                                                                                                                                                                                                                                                                                                    |  |
| <div><div>+Assignment(String title, String description, LocalDateTime openDate, LocalDateTime deadline)</div><div>-updateStatus()</div><div>+addSubmission(Submission submission)</div><div>+getSubmissions() : List&lt;Submission&gt;</div><div>+getSubmissionByStudent(String studentUsername) : Submission</div><div>+getTitle() : String</div><div>+setTitle(String title)</div><div>+getDescription() : String</div><div>+setDescription(String description)</div><div>+getStatus() : String</div><div>+isOpen() : boolean</div><div>+getTimeRemaining() : String</div><div>+hasStudentSubmitted(String username) : boolean</div></div> |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 226 647 812"> <p><b>Submission</b></p> <ul style="list-style-type: none"> <li>-String studentUsername</li> <li>-File file</li> <li>-LocalDateTime submissionTime</li> <li>-Integer grade</li> <li>-String feedback</li> </ul> <ul style="list-style-type: none"> <li>+Submission(String, File)</li> <li>+getGrade() : Integer</li> <li>+setGrade(Integer)</li> <li>+getFeedback() : String</li> <li>+setFeedback(String)</li> <li>+getFile() : File</li> <li>+getStudentUsername() : String</li> <li>+getSubmissionTime() : LocalDateTime</li> <li>+setFile(File)</li> <li>+setSubmissionTime(LocalDateTime)</li> </ul> </div> |  |
| <div data-bbox="208 868 642 1370"> <p><b>LecturerQuizzesPanel</b></p> <ul style="list-style-type: none"> <li>-DefaultListModel quizListModel</li> <li>-JList quizList</li> <li>-JButton addQuizButton</li> <li>-JButton editQuizButton</li> <li>-JButton deleteQuizButton</li> <li>-JButton viewScoresButton</li> </ul> <ul style="list-style-type: none"> <li>+LecturerQuizzesPanel()</li> <li>-addQuiz()</li> <li>-editQuiz()</li> <li>-deleteQuiz()</li> <li>-viewScores()</li> <li>-refreshQuizList()</li> </ul> </div>                                                                                                                        |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 229 750 724"> <div>StudentQuizzesPanel</div> <div> -DefaultListModel quizListModel<br/> -JList quizList<br/> -JButton startQuizButton<br/> -JButton viewQuizButton<br/> -List quizzes<br/> -String currentStudent<br/> -static Map&gt;&gt; studentAnswers<br/> -static Map&gt; submittedQuizzes </div> <div> +StudentQuizzesPanel(String)<br/> -openQuizForm(boolean)<br/> +static getStudentAnswers() : Map&gt;&gt;<br/> +static markQuizAsSubmitted(String, String)<br/> +static isQuizSubmitted(String, String) : boolean </div> </div> |  |
| <div data-bbox="208 791 860 1262"> <div>EditQuizDialog</div> <div> -JTextField quizTitleField<br/> -DefaultListModel&lt;String&gt; questionListModel<br/> -JList&lt;String&gt; questionList<br/> -Quiz quiz<br/> -DefaultListModel&lt;String&gt; quizListModel<br/> -int quizIndex </div> <div> +EditQuizDialog(Quiz quiz, DefaultListModel&lt;String&gt; qlm, int qi)<br/> -updateQuestionListDisplay()<br/> -editQuestion()<br/> -deleteQuestion()<br/> -addQuestion()<br/> -saveChanges() </div> </div>                                                     |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 228 1010 673"> <div> QuestionDialog </div> <div> -TextField questionTextField<br/> -DefaultListModel&lt;String&gt; optionsListModel<br/> -JList&lt;String&gt; optionsList<br/> -int correctAnswerIndex<br/> -Quiz quiz<br/> -Question question<br/> -int questionIndex<br/> -DefaultListModel&lt;String&gt; questionListModel<br/> -List&lt;String&gt; options </div> <div> +QuestionDialog(Quiz quiz, Question question, int questionIndex, DefaultListModel&lt;String&gt; questionListModel)<br/> +addOption()<br/> +editOption()<br/> +deleteOption()<br/> +setCorrectAnswer()<br/> +saveQuestion() </div> </div> |  |
| <div data-bbox="208 730 884 1158"> <div> QuizAnswerForm </div> <div> -String username<br/> -Quiz quiz<br/> -List&lt;Question&gt; questions<br/> -JPanel questionPanel<br/> -JButton submitButton<br/> -boolean isViewMode<br/> -MapUnsupported markdown: del&gt; studentQuizAnswers </div> <div> +QuizAnswerForm(String username, Quiz quiz, boolean isViewMode)<br/> -loadQuestions(List&lt;String&gt; answers)<br/> -submitAnswers()<br/> -calculateScore(List&lt;String&gt; answers) : int </div> </div>                                                                                                                              |  |

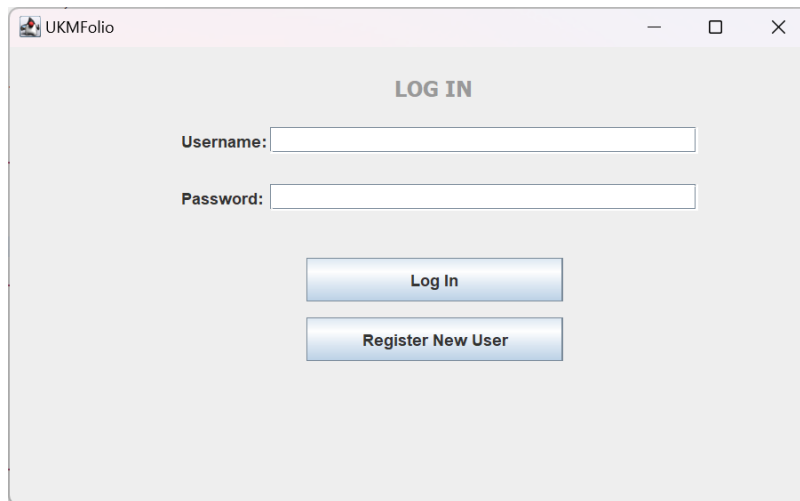
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 229 857 635"> <div>Question</div> <div>           -String text<br/>           -List&lt;String&gt; options<br/>           -int correctAnswerIndex         </div> <div>           +Question(String text, List&lt;String&gt; options, int correctAnswerIndex)<br/>           +setText(String text)<br/>           +getText() : String<br/>           +getOptions() : List&lt;String&gt;<br/>           +setOptions(List&lt;String&gt; options)<br/>           +getCorrectAnswerIndex() : int<br/>           +setCorrectAnswerIndex(int correctAnswerIndex)         </div> </div> |  |
| <div data-bbox="208 691 571 852"> <div>StudentScoresForm</div> <div></div> <div>+StudentScoresForm(Quiz)</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| <div data-bbox="208 906 571 1085"> <div>StudentMarks</div> <div>-static Map&lt;String, String&gt; studentMarks</div> <div>           +static storeMarks(String, String, int)<br/>           +static getMarks(String, String) : int         </div> </div>                                                                                                                                                                                                                                                                                                                                          |  |

|                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="208 226 533 585"> <p><b>Quiz</b></p> <p>-String title<br/>-List&lt;Question&gt; questions<br/>-static List&lt;Quiz&gt; quizList</p> <p>+Quiz(String t)<br/>+setTitle(String title)<br/>+getTitle() : String<br/>+getQuestions() : List&lt;Question&gt;<br/>+addQuestion(Question question)<br/>+removeQuestion(Question question)<br/>+static getQuizzes() : List&lt;Quiz&gt;</p> </div> |  |
| <div data-bbox="215 641 752 997"> <p><b>StudentAnnouncementsPanel</b></p> <p>-List instances<br/>-DefaultListModel announcementsListModel<br/>-JList announcementsList<br/>-JTextPane announcementDetails</p> <p>+StudentAnnouncementsPanel()<br/>-showSelectedAnnouncement()<br/>+refreshAnnouncements()<br/>+refreshAll()</p> </div>                                                                   |  |
| <div data-bbox="215 1053 831 1289"> <p><b>LecturerAnnouncementsPanel</b></p> <p>-DefaultListModel&lt;String&gt; announcementsListModel<br/>-JList&lt;String&gt; announcementsList</p> <p>+LecturerAnnouncementsPanel()<br/>-showAnnouncementDialog(Integer editIndex)</p> </div>                                                                                                                         |  |

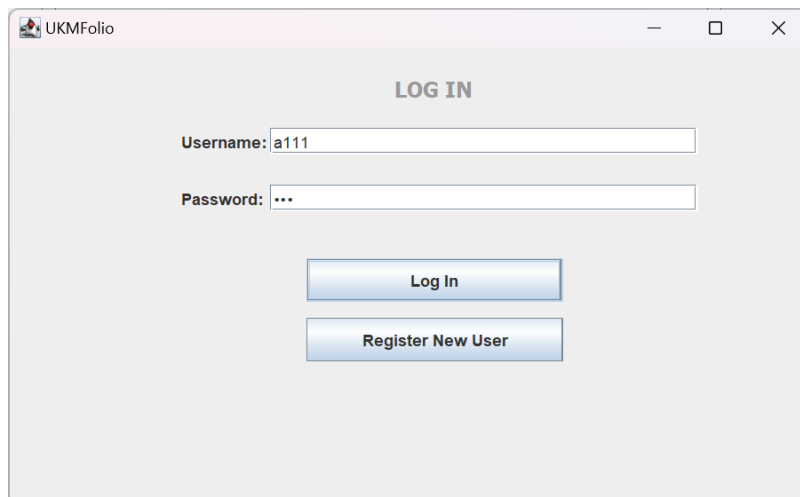
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <div data-bbox="215 228 799 552"> <div> <b>AnnouncementsManager</b> </div> <div>           -static AnnouncementsManager instance<br/>           -List&lt;String&gt; announcements         </div> <div>           -AnnouncementsManager()<br/>           +static getInstance() : AnnouncementsManager<br/>           +addAnnouncement(String announcement)<br/>           +updateAnnouncement(int index, String announcement)<br/>           +getAnnouncements() : List&lt;String&gt;         </div> </div> |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |

### 3.3 SAMPLE OUTPUT

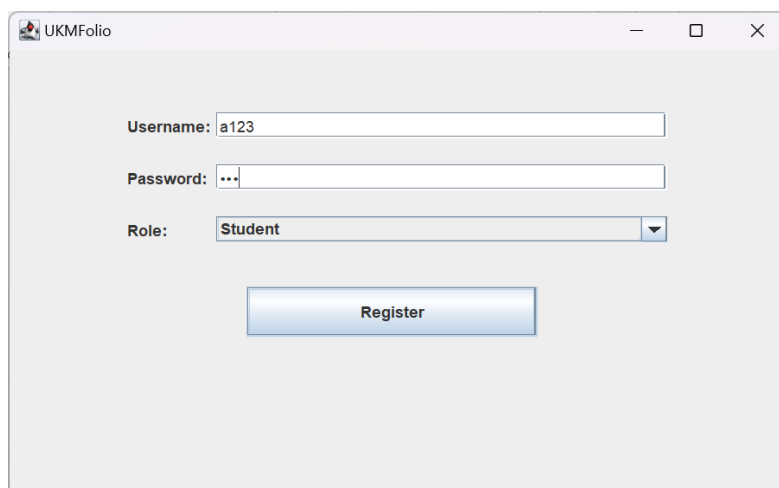
#### 1. User Log in & Register Interface



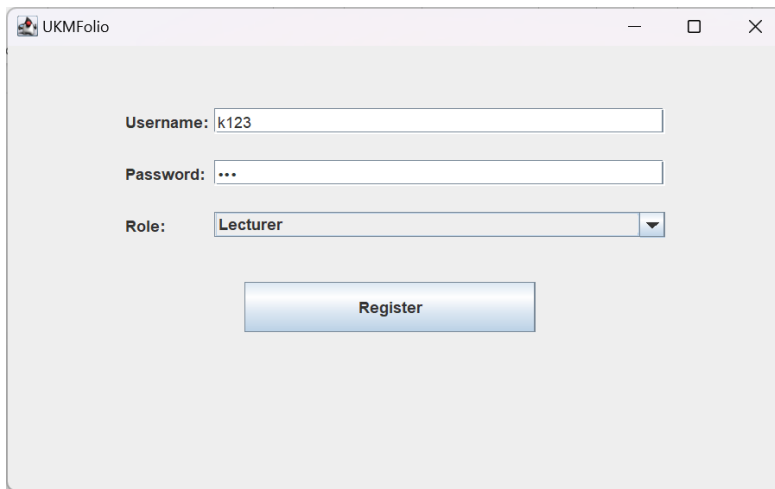
The screenshot shows a window titled "UKMFolio" with a light gray background. At the top center, the text "LOG IN" is displayed. Below it, there are two input fields: "Username:" followed by a text box and "Password:" followed by a text box. Below the password field, there are two buttons: "Log In" and "Register New User".



The screenshot shows the same "UKMFolio" window. The "LOG IN" text is still at the top. The "Username:" field now contains the text "a111". The "Password:" field is still empty, showing three dots. The "Log In" and "Register New User" buttons are still present below the fields.



The screenshot shows the same "UKMFolio" window. The "LOG IN" text is still at the top. The "Username:" field now contains the text "a123". The "Password:" field is still empty, showing three dots. Below the password field, there is a "Role:" label followed by a dropdown menu showing "Student". Below the dropdown menu, there is a single button labeled "Register".



UKMFolio

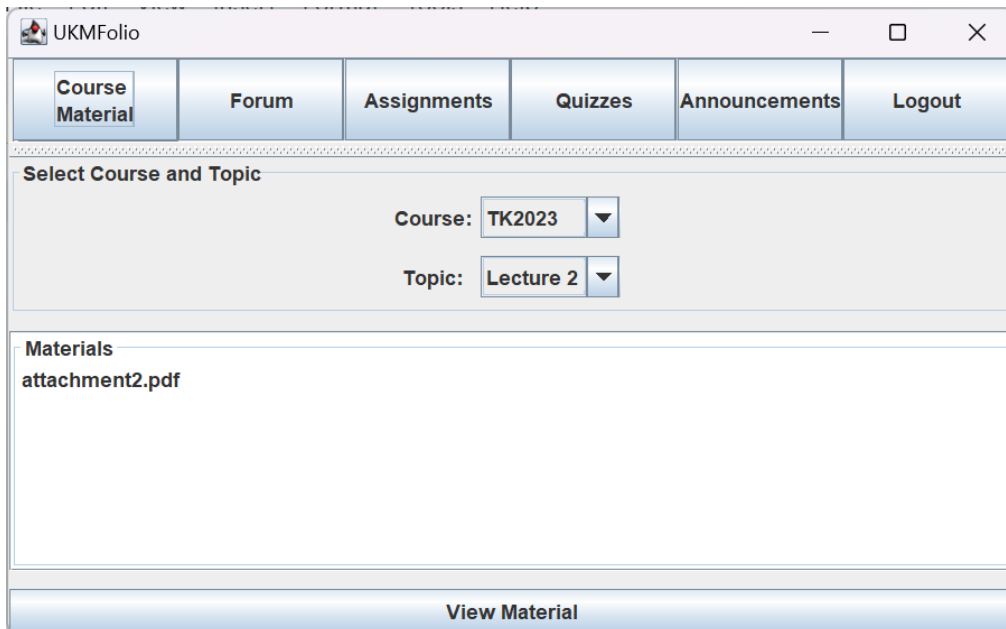
Username:

Password:

Role:

## 2. Course Material Interface

### a. Student's View



UKMFolio

|                        |              |                    |                |                      |               |
|------------------------|--------------|--------------------|----------------|----------------------|---------------|
| <b>Course Material</b> | <b>Forum</b> | <b>Assignments</b> | <b>Quizzes</b> | <b>Announcements</b> | <b>Logout</b> |
|------------------------|--------------|--------------------|----------------|----------------------|---------------|

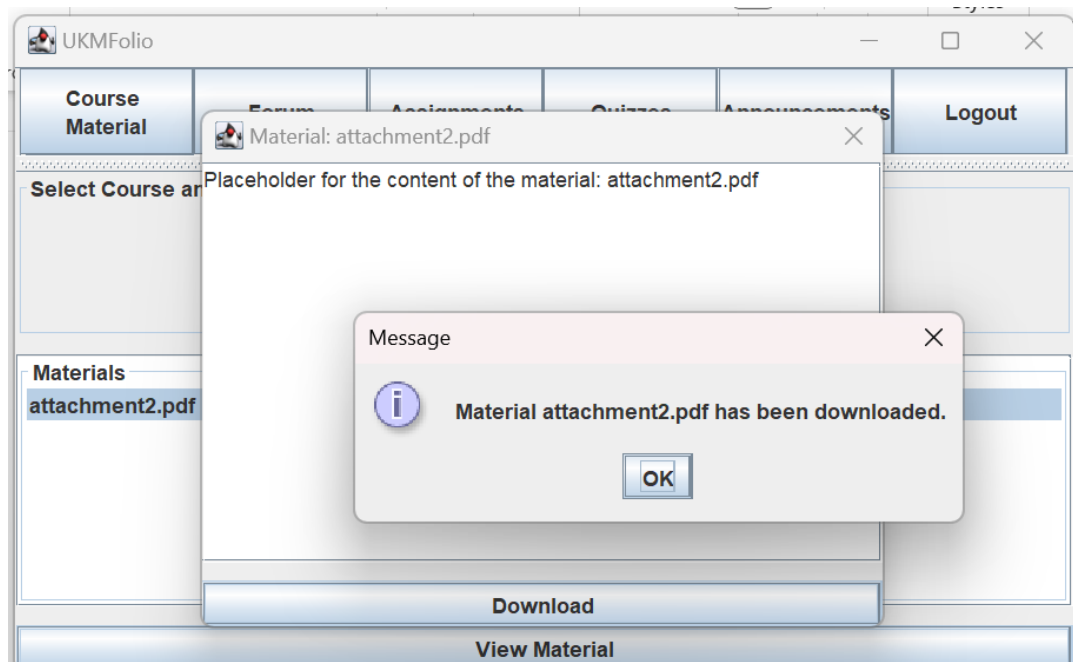
Select Course and Topic

Course:

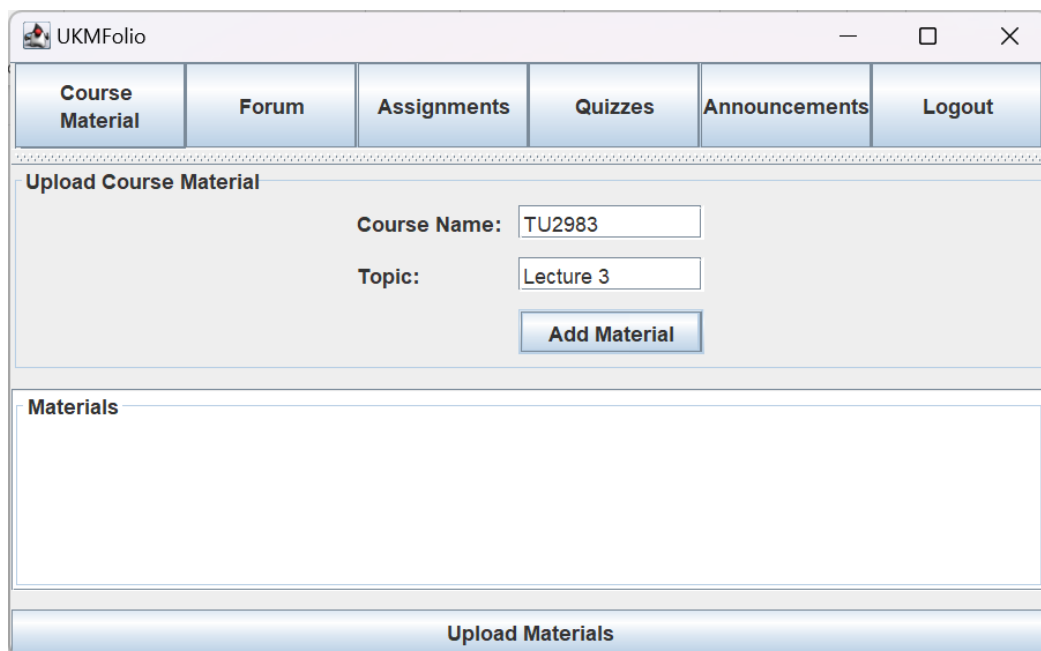
Topic:

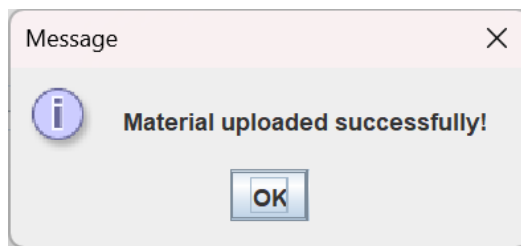
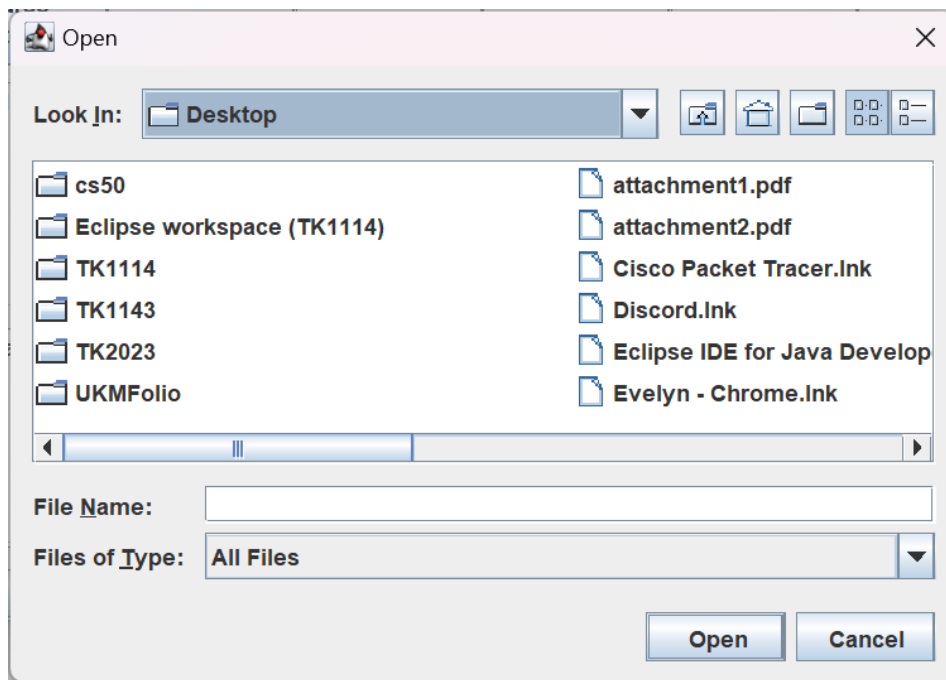
**Materials**

attachment2.pdf



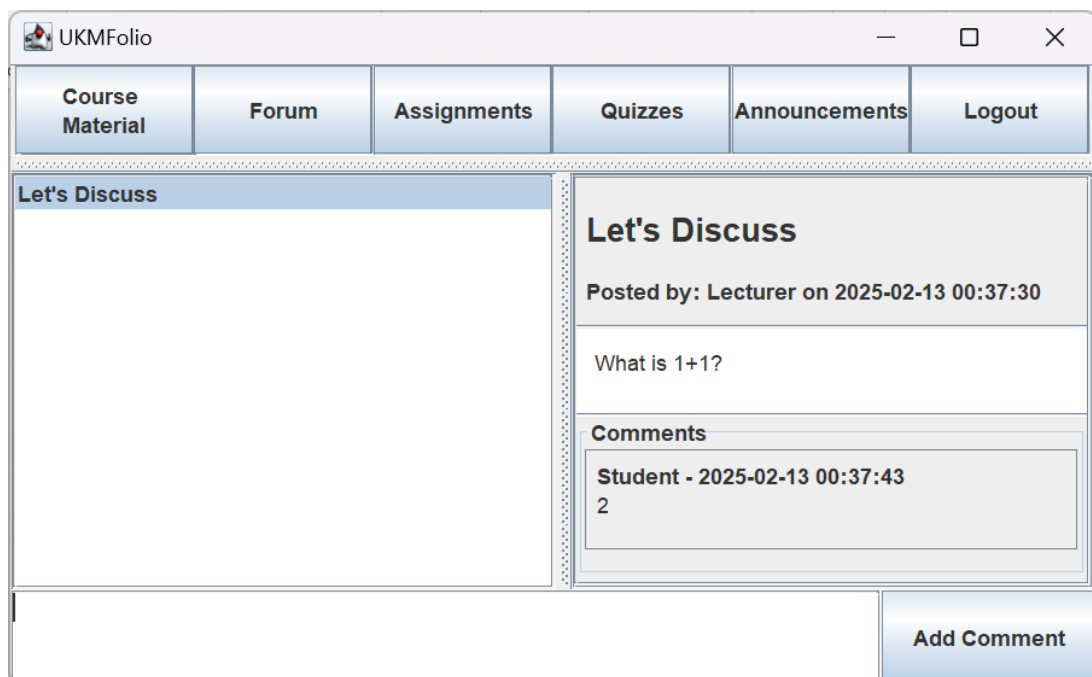
#### b. Lecturer's View





### 3. Forum Interface

#### a. Student's View



#### b. Lecturer's View

UKMFolio

Course Material
Forum
Assignments
Quizzes
Announcements
Logout

Title:

Create Discussion

Let's Discuss

### Let's Discuss

Posted by: Lecturer on 2025-02-13 00:37:30

What is 1+1?

**Comments**

Student - 2025-02-13 00:37:43

2

#### 4. Assignment Interface

##### a. Student's View

UKMFolio

Course Material
Forum
Assignments
Quizzes
Announcements
Logout

### Assignment Dashboard

| Title      | Description         | Open Date  | Due Date   | Status    | Time Re...  | Grade | Feedback  |
|------------|---------------------|------------|------------|-----------|-------------|-------|-----------|
| Tutorial 1 | submit 1 file only. | 2025-02... | 2025-02... | Submitted | 8 hours ... | 99    | Good job! |

Upload Submission
View Details
Select an assignment to view options

Assignment Details

Title:

Tutorial 1

Description:

submit 1 file only.

Open Date:

2025-02-13 00:38

Due Date:

2025-02-13 09:30

Status:

Open

Submission Date:

2025-02-13 00:38

Submitted File:

attachment2.pdf

Grade:

99

Feedback:

Good job!

Close

## b. Lecturer's View

UKMFolio

Course Material

Forum

Assignments

Quizzes

Announcements

Logout

Assignment Management

| Title      | Description         | Open Date        | Due Date         | Status | Submissions |
|------------|---------------------|------------------|------------------|--------|-------------|
| Tutorial 1 | submit 1 file only. | 2025-02-13 00:38 | 2025-02-13 09:30 | Open   | 1           |

Create Assignment

Grade Submission

View Submissions

| View Submissions - Tutorial 1 |                   |                  |                 |       |           |
|-------------------------------|-------------------|------------------|-----------------|-------|-----------|
| Student                       | Submission Status | Submission Time  | File            | Grade | Feedback  |
| a123                          | Submitted         | 2025-02-13 00:38 | attachment2.pdf | 99    | Good job! |
| a124                          | Not Submitted     | -                | -               | -     | -         |

View File
Close

File View - attachment2.pdf

Student: a123  
File: attachment2.pdf  
Submitted: 2025-02-13 00:38

Assignment Content

This is a placeholder for the actual file content.  
In a complete implementation, this would show the actual content of:  
attachment2.pdf

Close

Grade Submissions

a123

Grade (0-100):

100

Feedback:

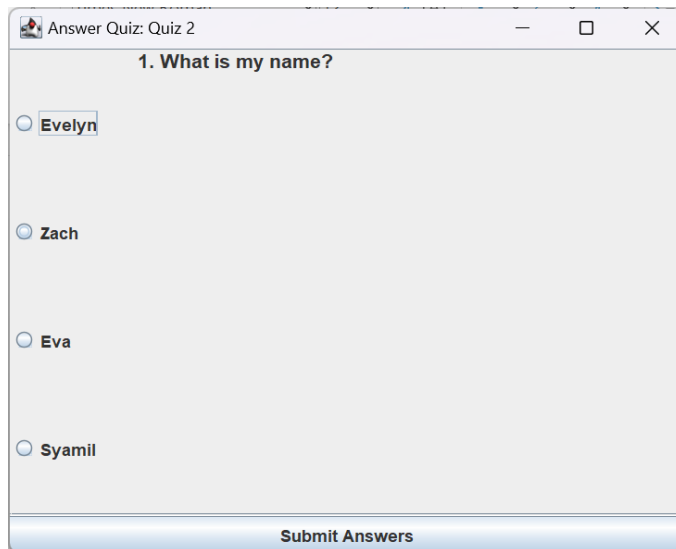
Well Done

Save Grade

Close

## 5. Quiz Interface

### a. Student's View



Answer Quiz: Quiz 2

1. What is my name?

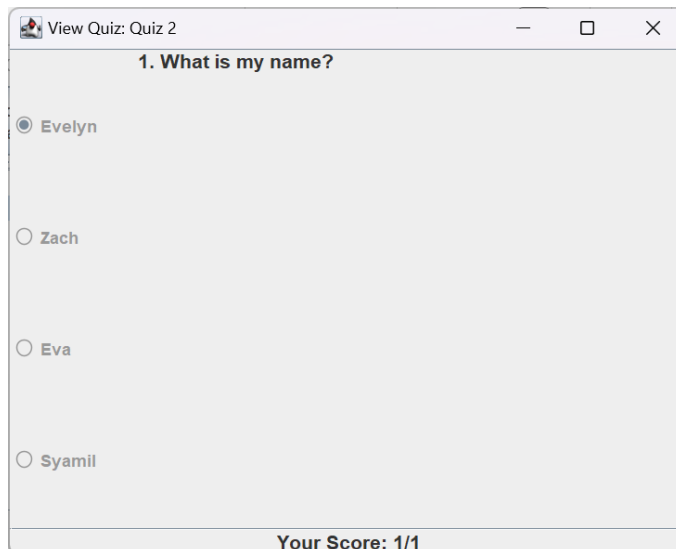
☒ Evelyn

☐ Zach

☐ Eva

☐ Syamil

Submit Answers



View Quiz: Quiz 2

1. What is my name?

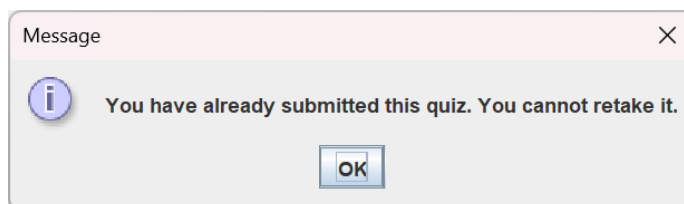
☒ Evelyn

☐ Zach

☐ Eva

☐ Syamil

Your Score: 1/1



Message

 You have already submitted this quiz. You cannot retake it.

OK

### b. Lecturer's View

Edit Quiz

Quiz Title: Quiz 2

What is my name?[Evelyn, Zach, Eva, Syamil]

Edit Question Delete Question Add Question

Add Question

Question: What is my name?

Evelyn  
Zach  
Eva  
Syamil

Add Option Edit Option  
Delete Option Set Correct Answer  
Save Question

Message

 Correct answer set: Evelyn

OK

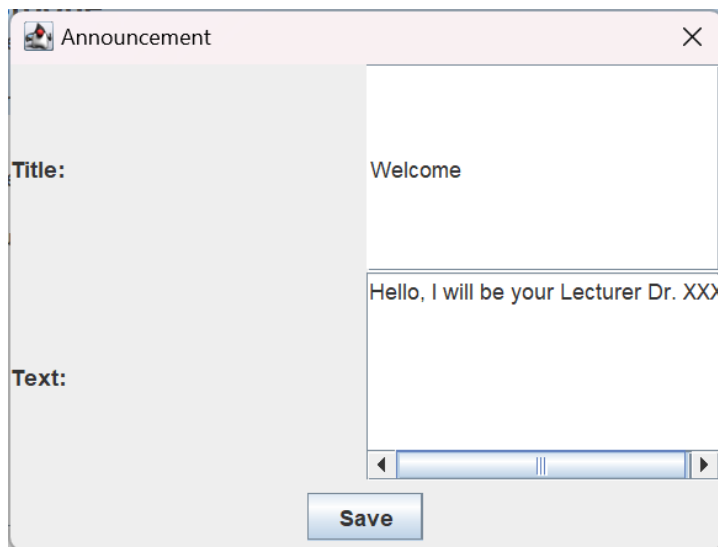
## 6. Announcements

### a. Student's View

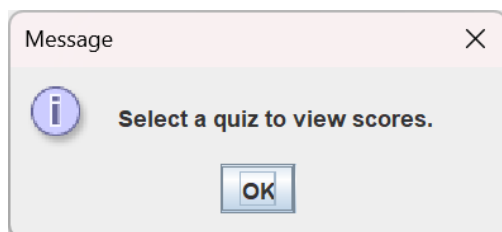
The screenshot shows a web browser window titled "UKMFolio" with a navigation bar containing links for Course Material, Forum, Assignments, Quizzes, Announcements, and Logout. The main content area displays two announcement cards side-by-side. The left card has a blue header "Lecture Postpone" and contains the text: "Posted by: Lecturer on 2025-02-13 00:43:15", "This week's lecture will be held online on Thursday 2pm.", "Welcome", "Posted by: Lecturer on 2025-02-13 00:43:41", and "Hello, I will be your lecturer for this semester, Dr XXXX." The right card has a white header "Lecture Postpone" and contains the text: "Posted by: Lecturer on 2025-02-13 00:43:15" and "This week's lecture will be held online on Thursday 2pm."

### b. Lecturer's View

The screenshot shows a web browser window titled "UKMFolio" with the same navigation bar as the student view. The main content area displays a single announcement card with a blue header "Lecture Postpone" and the text: "Posted by: Lecturer on 2025-02-13 00:43:15", "This week's lecture will be held online on Thursday 2pm.", "Welcome", "Posted by: Lecturer on 2025-02-13 00:43:41", and "Hello, I will be your lecturer for this semester, Dr XXXX." At the bottom of the interface, there are two buttons: "Add Announcement" and "Edit Announcement".



## 7. Error Handling



**Screenshots of system screens when the program is executed.**